

Experiment 14

Question: Experiment to understand feature scaling.

Aim

To understand feature scaling and apply Min-Max scaling, Standardization and Robust scaling.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_14():
    from sklearn.preprocessing import MinMaxScaler, StandardScaler, RobustScaler
    rng = np.random.default_rng(2)
    df = pd.DataFrame({"Age": rng.integers(20, 60, 100),
                       "Salary": rng.normal(50000, 15000, 100).round(0),
                       "Experience": rng.integers(0, 30, 100)})
    print("Original statistics:\n", df.describe().loc[["mean", "std", "min", "max"]].round(2),
          "\n")

    mm = pd.DataFrame(MinMaxScaler().fit_transform(df), columns=df.columns)
    ss = pd.DataFrame(StandardScaler().fit_transform(df), columns=df.columns)
    rb = pd.DataFrame(RobustScaler().fit_transform(df), columns=df.columns)

    print("Min-Max scaling (x - min) / (max - min):\n",
          mm.describe().loc[["mean", "std", "min", "max"]].round(3), "\n")
    print("Standardization (x - mean) / std:\n",
          ss.describe().loc[["mean", "std", "min", "max"]].round(3), "\n")

    # Manual calculation check for the first Salary value
    x = df["Salary"].iloc[0]
    print("Manual check for first salary", x)
    print("Min-Max :", round((x - df["Salary"].min()) / (df["Salary"].max() -
    df["Salary"].min()), 4),
          "| sklearn :", round(mm["Salary"].iloc[0], 4))
    print("Z-score :", round((x - df["Salary"].mean()) / df["Salary"].std(ddof=0), 4),
          "| sklearn :", round(ss["Salary"].iloc[0], 4))

    fig, ax = plt.subplots(1, 4, figsize=(16, 3.8))
    for a, d, t in zip(ax, [df, mm, ss, rb], ["Original", "Min-Max", "Standardized", "Robust"]):
        for c in d.columns:
            sns.kdeplot(d[c], ax=a, label=c)
        a.set_title(t)
    ax[0].legend()
    plt.tight_layout()
    plt.show()
```

exp_14()

Output

Original statistics:

	Age	Salary	Experience
mean	40.16	50265.78	14.99
std	10.77	14382.15	8.64
min	21.00	13492.00	0.00
max	59.00	86860.00	29.00

Min-Max scaling (x - min) / (max - min):

	Age	Salary	Experience
mean	0.504	0.501	0.517
std	0.283	0.196	0.298
min	0.000	0.000	0.000
max	1.000	1.000	1.000

Standardization (x - mean) / std:

	Age	Salary	Experience
mean	0.000	0.000	-0.000
std	1.005	1.005	1.005
min	-1.788	-2.570	-1.744

max 1.758 2.557 1.630

Manual check for first salary 65523.0

Min-Max : 0.7092 | sklearn : 0.7092

Z-score : 1.0662 | sklearn : 1.0662

Graph

